

HAL Campus Evidence Desk

Private, auditable AI for campus operations and research.

Supplied evidence in -> constrained synthesis -> deterministic acceptance -> accountable human review.

HAL SUPREME / Jakob Hedrich

Campus decisions are evidence-heavy, but evidence is fragmented.

Today

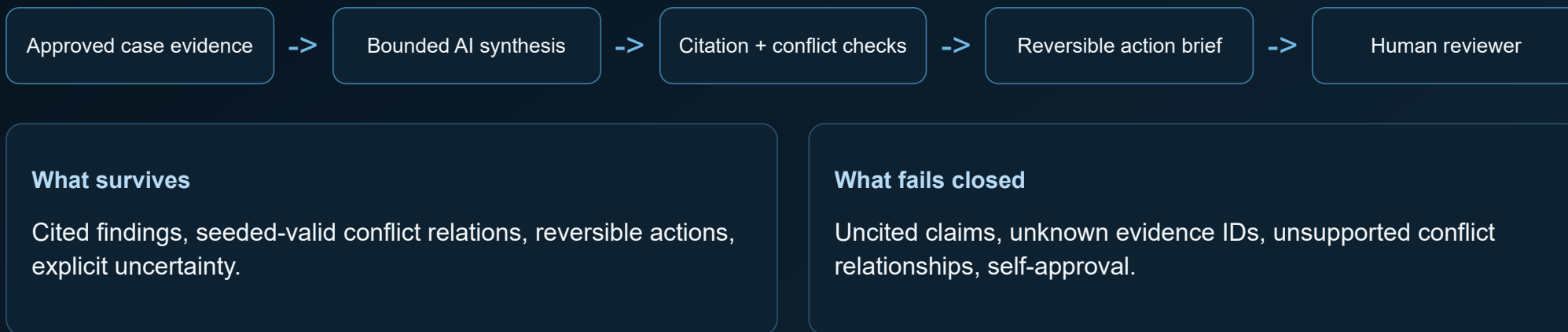
- Policies, requests, incident notes, research records, and facilities data live in separate systems.
- Staff manually reconcile sources before taking action.
- Contradictions can be missed or resolved inconsistently.

Generic AI risk

- Fluent output may omit provenance.
- Inference can look like fact.
- Unsupported conflicts can be asserted confidently.
- Institutional authority can become unclear.

The need is not merely a smarter chatbot. It is an auditable evidence contract around AI assistance.

Evidence-scoped synthesis with a mandatory human-review gate.



Three canonical campus scenarios.

Conflicting policy guidance

Detect incompatible after-hours approval paths and pause for policy-owner resolution.

Research access request

Identify missing owner approval, purpose, retention, and secure workspace requirements without inventing a conflict.

Facilities energy anomaly

Prioritize a timed manual air-handler override before weaker occupancy or weather explanations.

Reviewer experience

Every accepted item exposes source IDs. The report remains **PENDING_HUMAN_REVIEW**.

04 - VALIDATION

The system was tested against both citation errors and relational hallucination.

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automated tests passing

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bounded live NVIDIA NIM cases completed without provider error

100%

deterministic citation validity across canonical cases

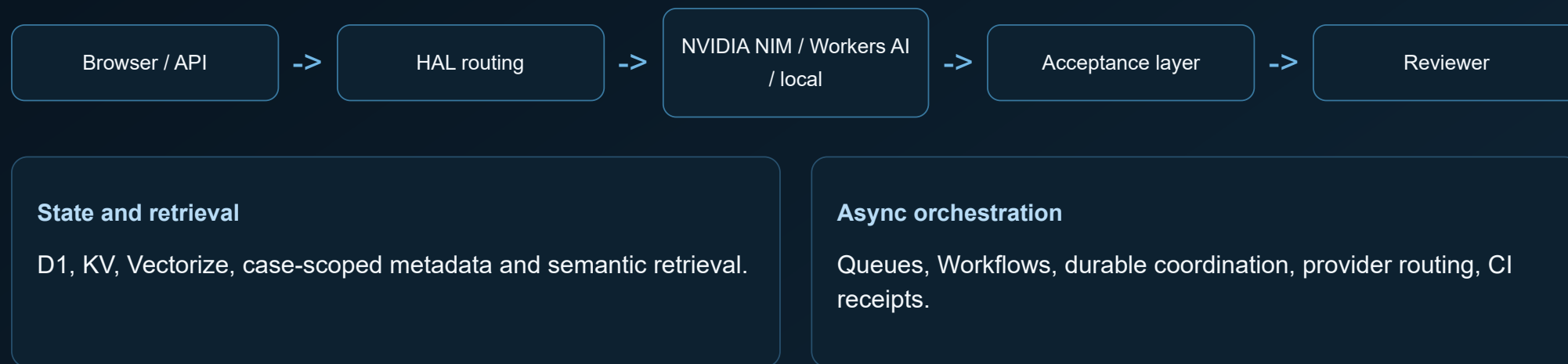
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unsupported material claims in deterministic canonical reports

A live validation pass exposed a subtle failure mode: a model could cite valid evidence IDs yet invent a false relationship between them. The acceptance layer now rejects unsupported conflict relations, and a regression test locks that behavior in.

Live validation: 1,409 total tokens; mean latency about 18.6 seconds in the current prototype path.

Provider-independent AI, deterministic acceptance, modular infrastructure.



Model providers are replaceable. The evidence acceptance policy is not.

Start with one bounded workflow. Measure it. Expand only after evidence.

Pilot metrics

- Reviewer time-to-source
- Citation validity
- Unsupported-claim rate
- Contradiction detection
- Time saved preparing briefs
- Latency and cost per path

Scale-up sequence

1. One approved workflow
2. SSO + audit + retention
3. Institution connectors
4. Local/open provider options
5. Additional workflows only after the same acceptance gates pass

Business-model hypothesis: institution-hosted or managed deployment, with a free/open public-interest core where practical and paid integration/support/managed infrastructure where needed.

AI assistance institutions can inspect, challenge, and govern.

The competitive advantage is not model size. It is the discipline around the model: provenance, bounded synthesis, deterministic acceptance, explicit uncertainty, and accountable human review.

Next milestone: a bounded institutional workflow pilot with measurable review-time and evidence-quality outcomes.

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